

Siemens : S7-Functions protocol driver

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Overview

The S7 Functions driver uses built-in S7 function blocks to communicate with the S7 via L2/MPI, ISO Industrial Ethernet or TCP/IP (if supported on your CP). You will need to obtain “S7 Functions” or S7 drivers + licences for your CP card from SIEMENS to enable communication.

Communication to H1 BUS using S7 functions for H1 (to a CP 443-1) has only been tested for Industrial Ethernet, and not for TCP/IP.

This document will take you through the installation steps for SIMATIC NET, and then of the Adroit driver. Depending on your version of Simatic and which CP card you use, instructions will differ. **The driver will not appear in the available list in Adroit Setup until you have completed installation of the Simatic net drivers.**

Wiring

L2: Wiring between the card and the stations is standard RS-485. Fibre optic link modules (OLM's) and copper may be used. No external terminators as such are required, however it is necessary to set up terminating switches at the end of cable runs. As physical wiring is not within the scope of this document, please consult documentation accompanying the communication modules for details.

H1: Use an AUI to BNC /UTP /Fibre converter to connect to the LAN.

Hardware installation

When using Advanced PC configuration package, please follow the automatic installation process in the SIMATIC manuals.

CP5412-A2

Unless the card is already successfully installed and working in the host machine the following steps must be followed. If the card is already installed, determine the I/O port and IRQ used and then follow the instructions in the *Sinec Driver installation and setup* section after removing or disabling the existing drivers.

1. Click on Start->Programs->Administrative Tools->Windows NT Diagnostics.
2. The “Windows NT Diagnostics” program will run.
3. Click on the Resources tab in the dialog that appears.
4. Find a free IRQ, it must be 5,10,11,12 or 15. If un-sure use IRQ 12 or 11 if IRQ 12 is un-available.
5. Write it down someplace
6. Click on the button labeled “I/O Port” on the table, and check that addresses 240 to 243 are free. If not you will have to find another free address from the table overleaf.
7. Set the dip-switches on the card to indicate the free address.

0000	240-243 (default)
0001	244-247
0010	248-24B
0011	24C-24F
0100	280-283
0101	284-287
0110	288-28B
0111	28C-28F
1000	300-303
1001	304-307
1010	308-30B
1011	30C-30F
1100	390-393
1101	394-397
1110	398-39B
1111	39C-39F

The standard setting is “0000” with all switches off or down.

8. Shut down the computer and install the card in a free slot.
9. Switch on the computer.
10. If any problems are experienced during starting up you may have to change the I/O address.

Disabling an existing L2 driver

If any other NT device driver is already using the CP5412 card, it must be disabled or uninstalled first, skip this procedure if installing the card for the first time in this machine.

To use existing device agents with the new driver, you will have to delete the old devices in adroit driver setup, and then create them again after the new driver has been installed. Unless you wish to create different devices for use with the new driver delete all of the devices in setup before continuing. Devices can only be deleted with the driver that made them.

1. Open adroit set-up
2. Click on Drivers tab

3. Make a note of all driver settings for each device. (you could back up the registry key using regedit).
4. Delete each device

The following steps can be followed to disable the existing NT device driver.

1. Open Control Panel.
2. Double click on the Devices applet.
3. Find and select the old driver.
4. Then click on Startup and change it to disabled.
5. Reboot for the change to take effect.

Unless the existing driver has an uninstaller, do not remove any files until you are sure they are not needed.

Hardware installation: Ethernet

You will need to install an Ethernet card in your PC with the relevant driver, you can then add NetBEUI or other protocols as normal.

When installing the H1 component from the CD, you will be prompted to install the SINEC driver called “SIEMENS Industrial Ethernet (ISO)” from the Simatic net CD, this driver must then bind to your network card which will connect to the PLC.

Otherwise follow similar instructions as for L2.

Hardware installation: CP5613

1. Install the SIMATIC NET software components first
2. Shut down, and insert the CP card.
3. Upon start-up, Windows 2000 Professional will find the PCI P&P card, and install the card-drivers for you from the CD.
4. Open the applet called “Setting up the PC-PG Interface” from control panel; verify card is installed OK.
5. Continue below.

Driver installation and setup

Installation overview

The software installation is divided into 2 parts. The Sinec software must be installed first, followed by the Adroit driver installation. 3 different versions of Siemens software are catered for.

Sinec Software Installation

If using the [Advanced PC Configuration](#) disk please jump to that section below.

If using **S7-RedConnect** please jump to the [S7-Redconnect install procedure](#) section below.

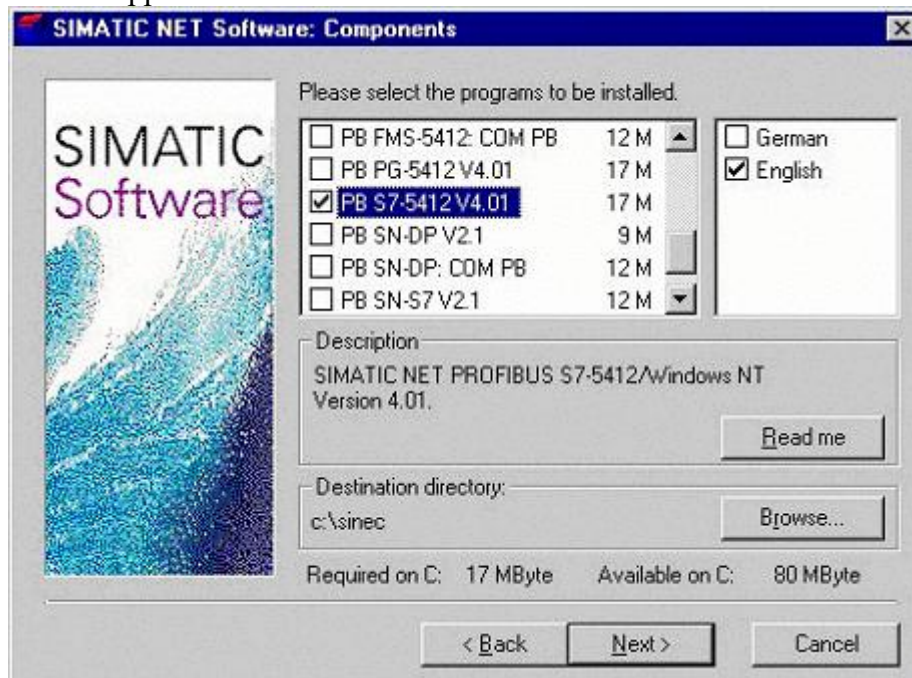
Before you will be able to install the SIES7FN driver in Adroit, you must install the Relevant Simatic Net Driver software and transfer the license from the key disk supplied. H1 users must install the Industrial Ethernet component here instead. If you enter the Adroit Setup\Drivers option without installing the S7 WinNT drivers first, you will NOT see the SIES7FN driver appear in the list of available drivers. If using any other CP, install the relevant components.

For example using a CP5613, we installed

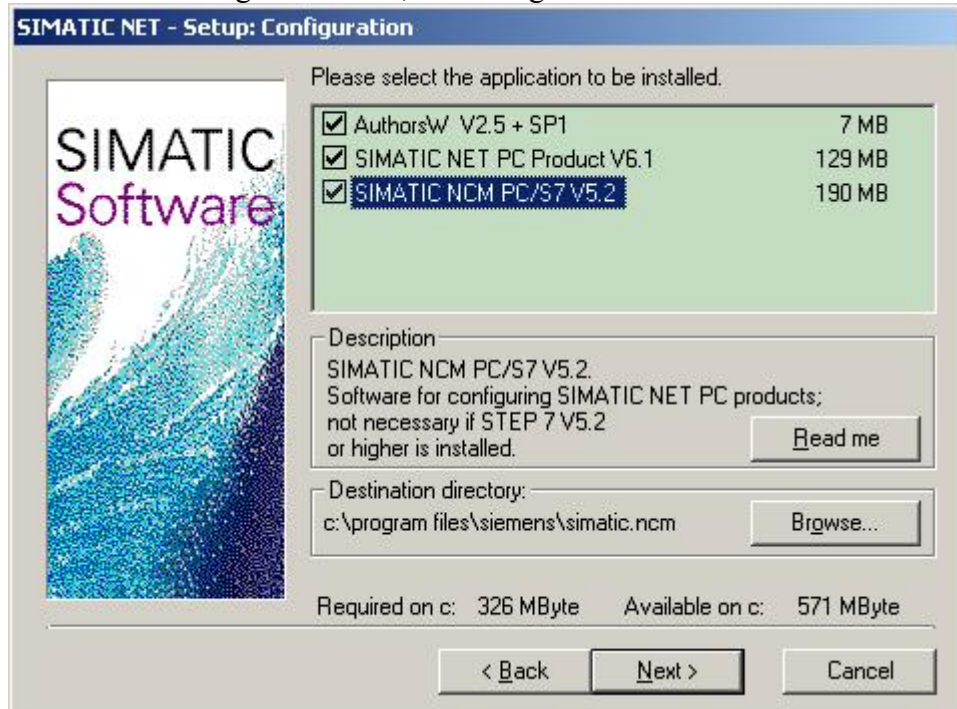
SIMATIC AuthorsW V2.3
SIMATIC NET S7-1613 V2.1
SIMATIC NET FMS-OPC-Server V2.2
SIMATIC NET S7/SR-OPC-Server V2.2
SIMATIC NET IE SOFTNET-S7 V3.3
SIMATIC NET CP 5613/CP 5614 V2.1
SIMATIC NET S7-5613 V2.1
SIMATIC NET PB SOFTNET S7 V5.3
Applied → Servicepack 2 Build 1407

If setting up with version 4 off the CD, installation details will differ. Alternate instructions appear where necessary. Version 4 only differs in the way the software is installed and what the configuration screens look like. The Adroit driver operation is identical for both versions.

1. To install the S7 WinNT drivers, run Setup on the installation disks provided.
2. Install the Siemens software to its default location "C:\SINEC"
3. The Version 2.0 Sinec setup program will display a page with two tick-boxes just before it finishes. Both of these must be checked as on (to automatically install the authorization and view the readme file).
 - If setting up with version 4 off the CD, the following dialog will appear make the selection as shown.



- If installing version 6.x, the dialog looks like

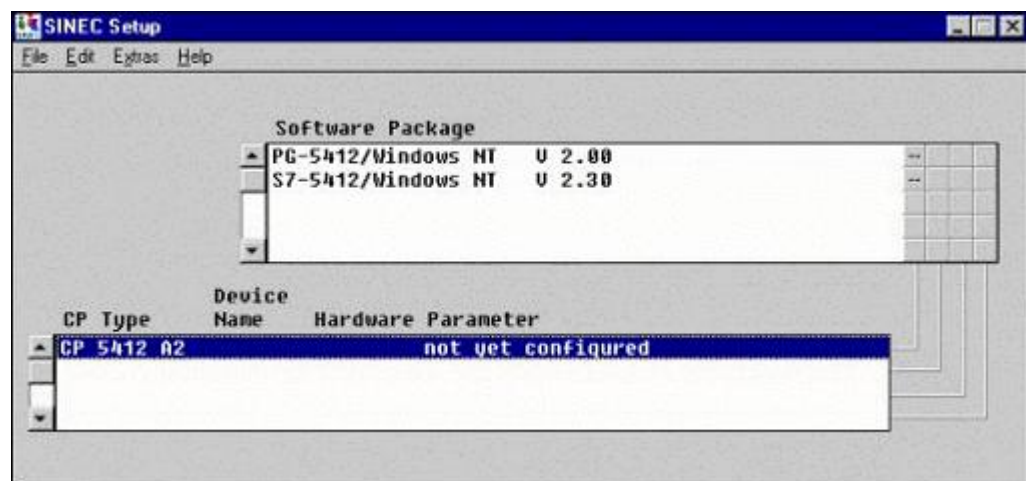


Ensure that the optional component “Simatic NCM PC/S7” is selected as well.

4. The information in the readme.txt file covers possible installation problems or limitations. The “Install License” option will transfer the license from the orange disk to your hard drive. Do not interrupt this procedure, or your license may be lost. You may choose to install/uninstall the license later if you wish using the Sinec setup program.

Simatic net version 2 instructions:

5. When you select the configuration option you will see an un-configured device as follows.



6. Select this device and then select **E**dit**M**odify configured CP from the menu, in order to configure the CP.

7. The dialogue screenshot shown below is used to set up your CP 5412-A2
8. Clicking on the checkbox labeled “S7-5412” should enable the S7-5412 software. Select the file wincc.ldb the first time you use this dialog, since the “ADROIT.LDB” file will not yet exist.
9. The Bus parameter calculation method selected should be the same as the PLC configured setting. Although either will work, it is strongly recommended that the same bus settings be used on all devices connected to the L2 network.
10. Select I/O port, interrupt and dual port RAM addresses, taking existing hardware into account to avoid hardware clashes. To change these settings if you are using version 4 or later, see instructions in the section **“Changing hardware settings”** below.
11. For compatible databases, choose WINCC.LDB in SINEC\COMS7.NT as the default. Later in the installation you can change this to ADROIT.LDB.
12. Fill in the other L2 bus settings as per screenshot.

SINEC Reconfiguration CP 5412 (A2)

PC hardware parameters

Device name: Interrupt:

I/O port: Dual-port RAM:

Compatible software packages on CP 5412 (A2)

Protocol: ☒ S7-5412
☐ FMS (+FDL)-5412
☐ DP (+FDL)-5412
☐ PG-5412
☐ Scope L2

Compatible databases:
 Selection
 Selection

☒ PROFIBUS ☐ MPI
☐ Time (Pilot)

L2 bus parameters

L2 address: Mode for L2 bus parameter calculation:

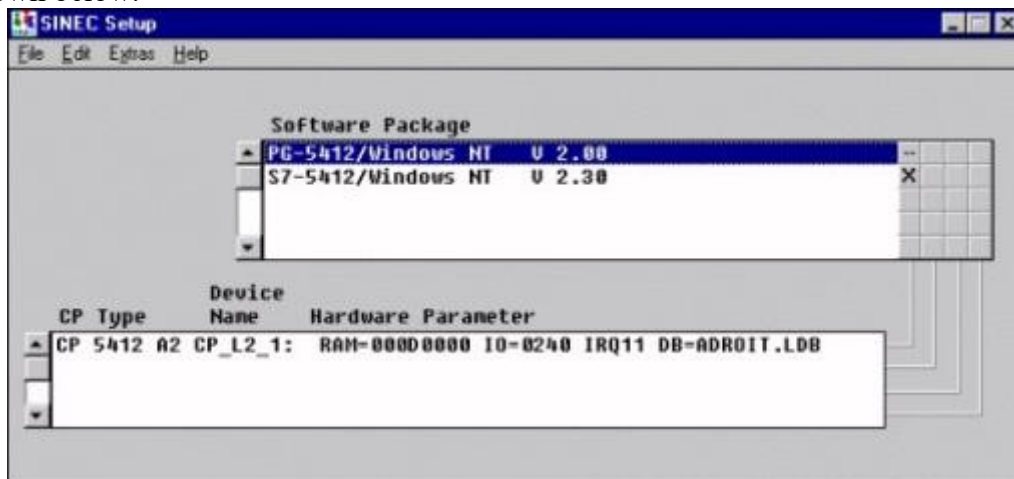
Highest station address:

Number of masters:

Number of slaves:

Transmission rate:

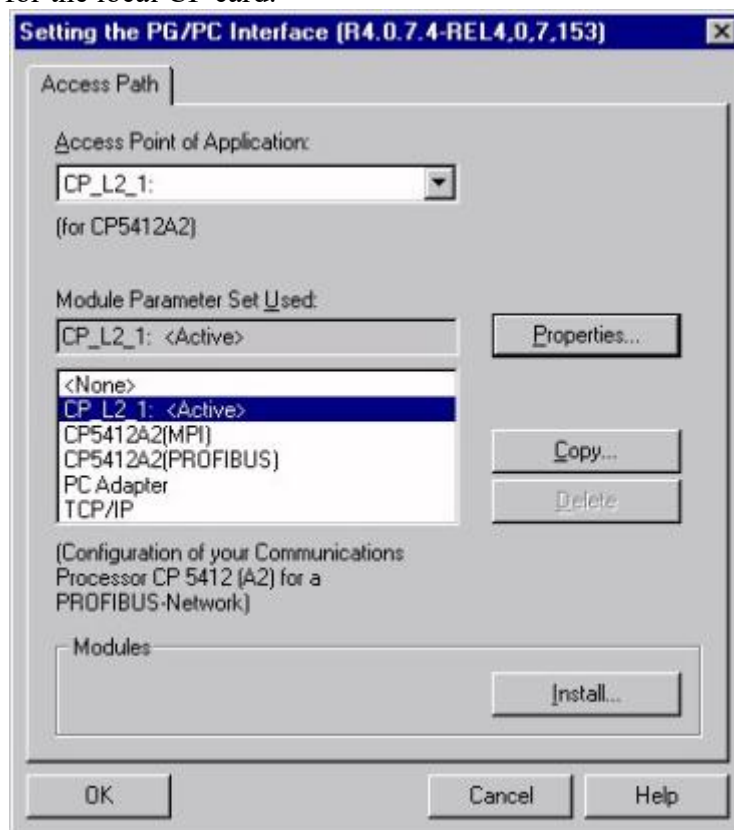
Once you accept this configuration, the SINEC setup dialogue will look similar to that shown below.



Simatic net version 4 installation instructions:

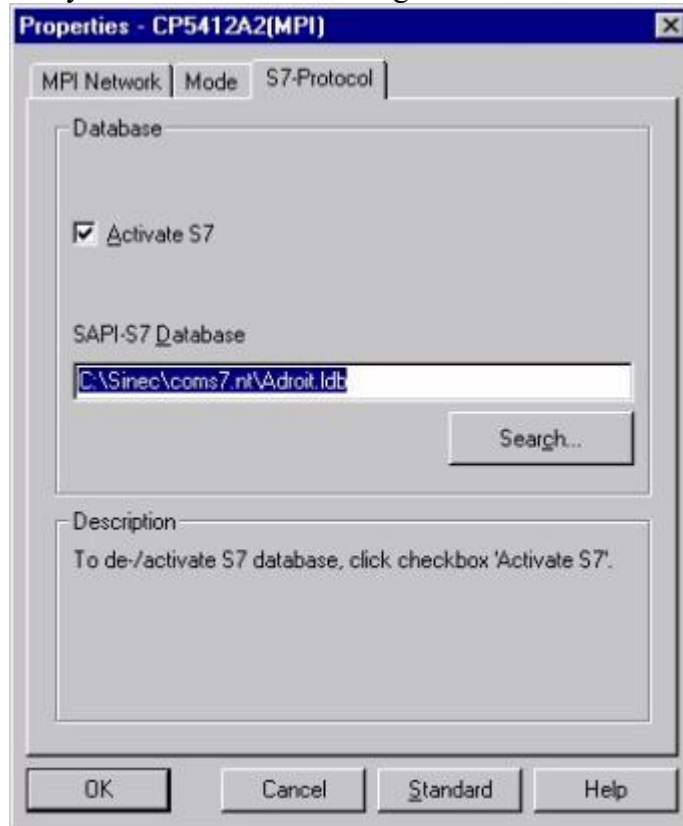
When the installation completes you should re-boot your computer.

5. Run the application "Setting the PG-PC interface.
6. Select the Access point of Application, this is the name for the device driver for the local CP card.



7. Select the communications option (MPI or PROFIBUS). H1 users must select "OSI Ind. Ethernet".
8. Click on properties to configure the parameter set. (see below)
9. Click on the checkbox to enable the S7 functions software, select the wincc.ldb file. (After you have configured the adroit driver, a new selection becomes available here, then you can select the adroit.ldb file.)
10. Click on the Mode page.

11. You may now Reset and Restart the CP with the new settings.
12. When you return to the selection screen, the <active> indicator will have moved to your selection if the configuration was successful.

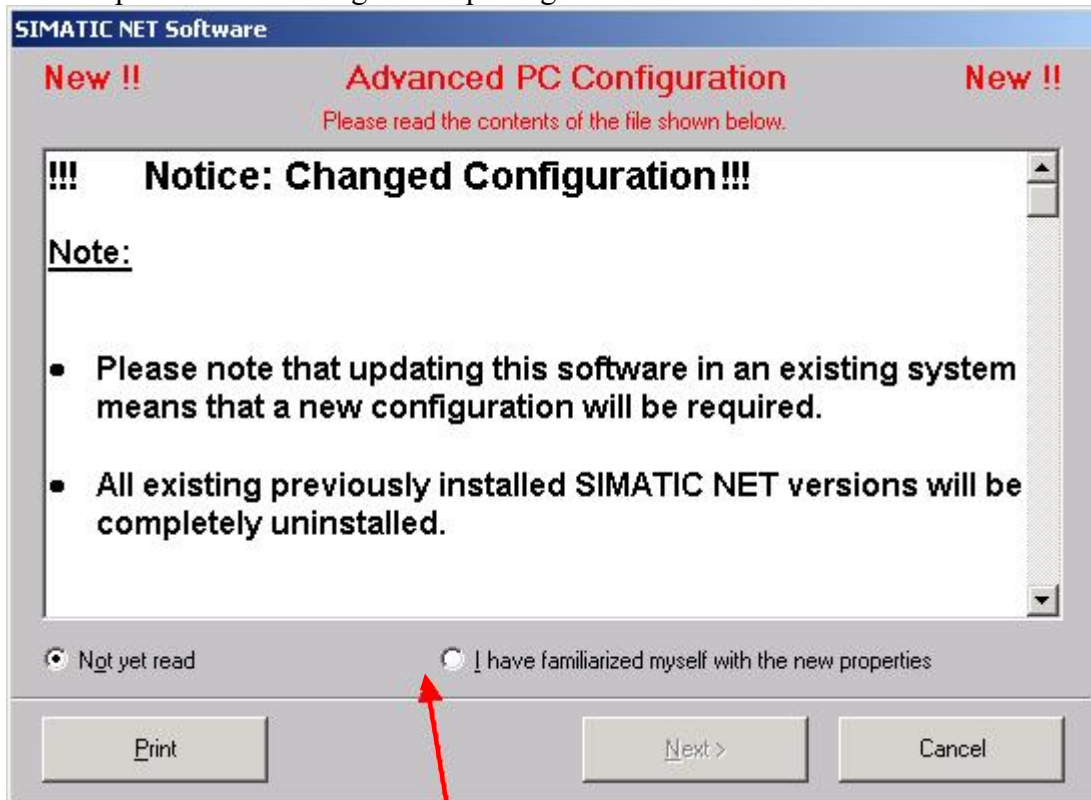
**General instructions:**

1. Re-boot the computer. This must always be done after changing any settings shown in the previous steps.
2. Find the Simatic program group in the root of your start menu.
3. Run the Program "Setup" in the SIMATIC NET program group.
4. Run the Adroit Setup Program and install the S7 Functions Driver.
5. You may now add an S7 device, as you will now be able to see the driver in the list of available drivers.
6. Once you have added at least one device by following the section "Device Configuration" below, go back to the SINEC configuration program and you will be able to specify ADROIT.LDB as your compatible database instead of WINCC.LDB. I.E. configure one Adroit driver device, then perform only step 11 (step 8 if using version 4) above using ADROIT.LDB.
7. **Re-boot the computer.**

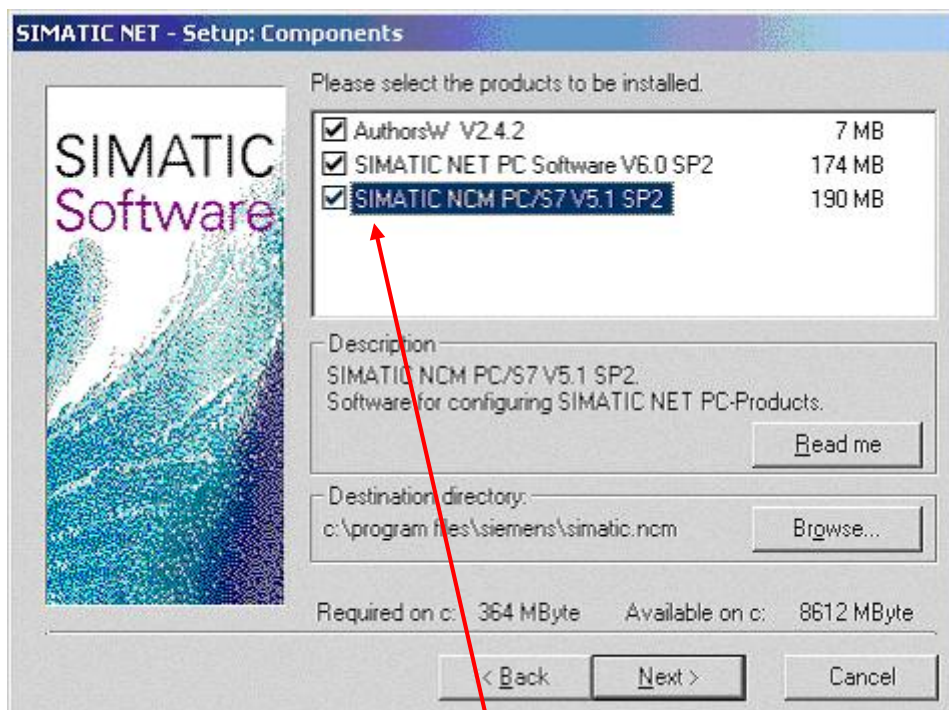
Advanced PC Configuration

The "Advanced PC Configuration" software is structured to work together with your Programming unit's Step-7 project. Users familiar with Step-7 projects will be familiar with the idea of downloading configuration to a PLC. In this case, configuration must be "downloaded" to the PC, every time you change a connection of other setting, the configuration has to be compiled and downloaded again. Step 7 is not required.

The setup for the PC Configuration package starts with



Click the radio button to continue.

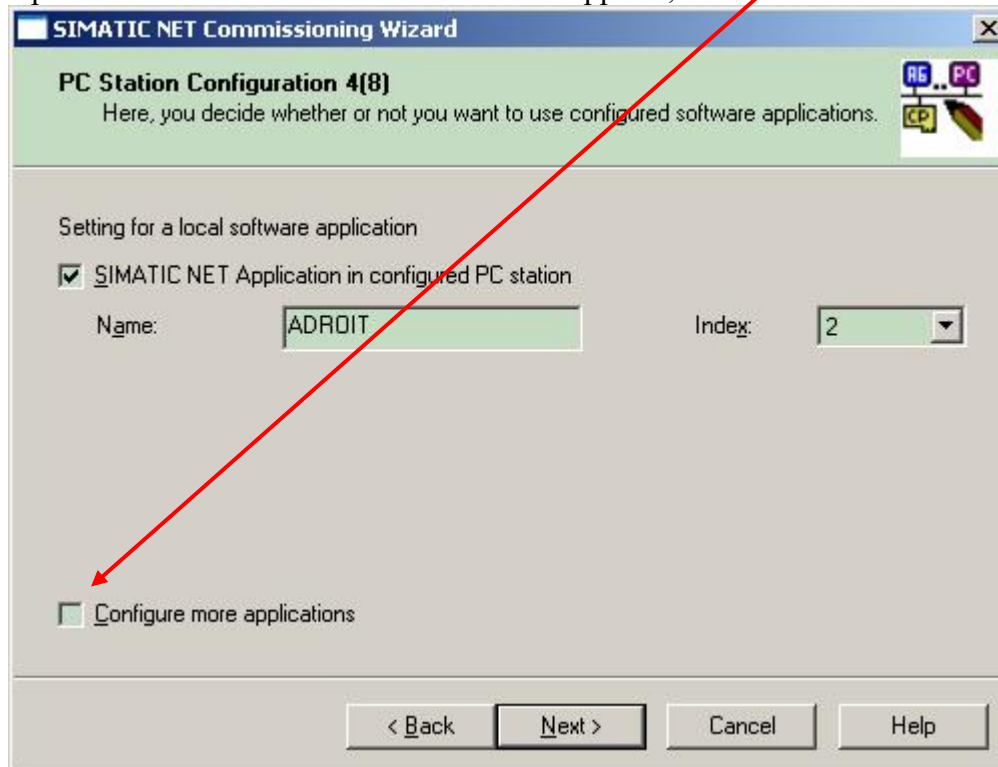


To install, simply select all three components.

This driver uses the NCM component.... to perform configuration tasks.

NOTE: If you are using Step 7 on this computer, you cannot install the NCM component as well. Please use Step 7 to configure and download the connections, the configuration process is similar to that described below, except in how the Wizard is invoked, you will not have the PC configuration Wizard.

Optional : In the Hardware wizard that first appears, select the bottom tick-box,

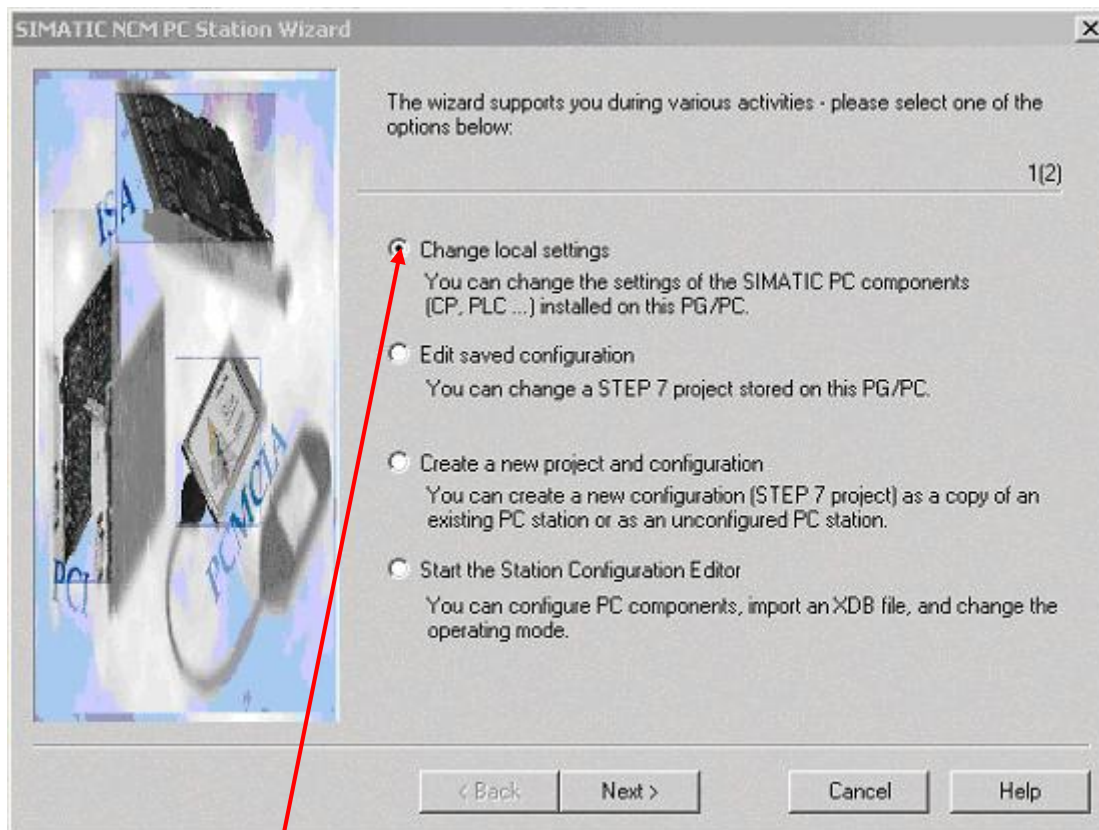


and add ADROIT as an application on the next wizard step, (*show above*) leave it blank in order to end the wizard. This step may be performed later, but it helps to do it right now.

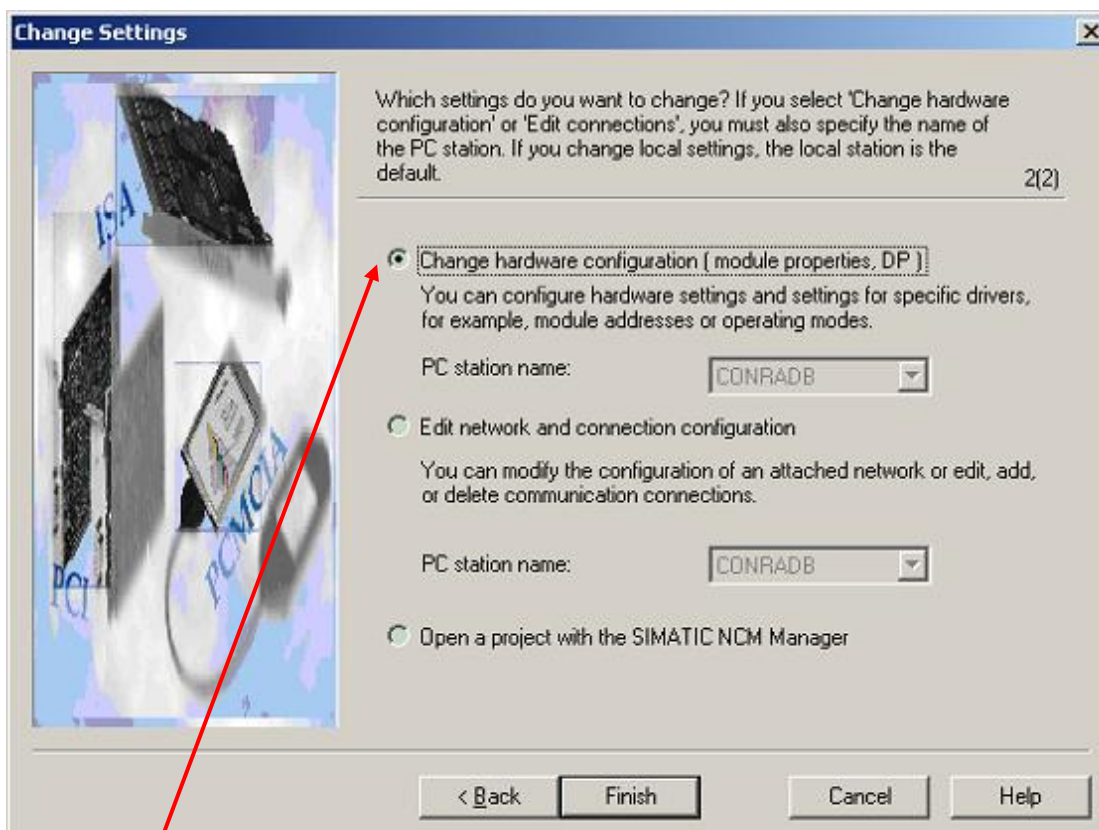
To set up the PC

Start the PC Configuration wizard from the desktop icon or shortcut in Program Files/SIMATIC...

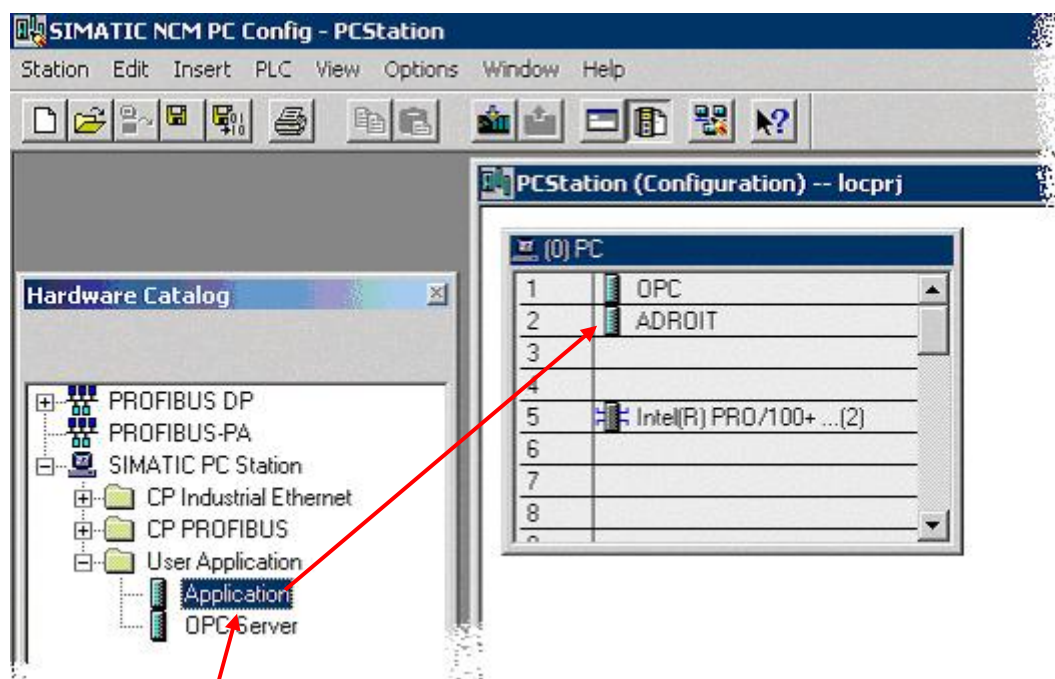




Select “Change local settings”, then click “Next”.

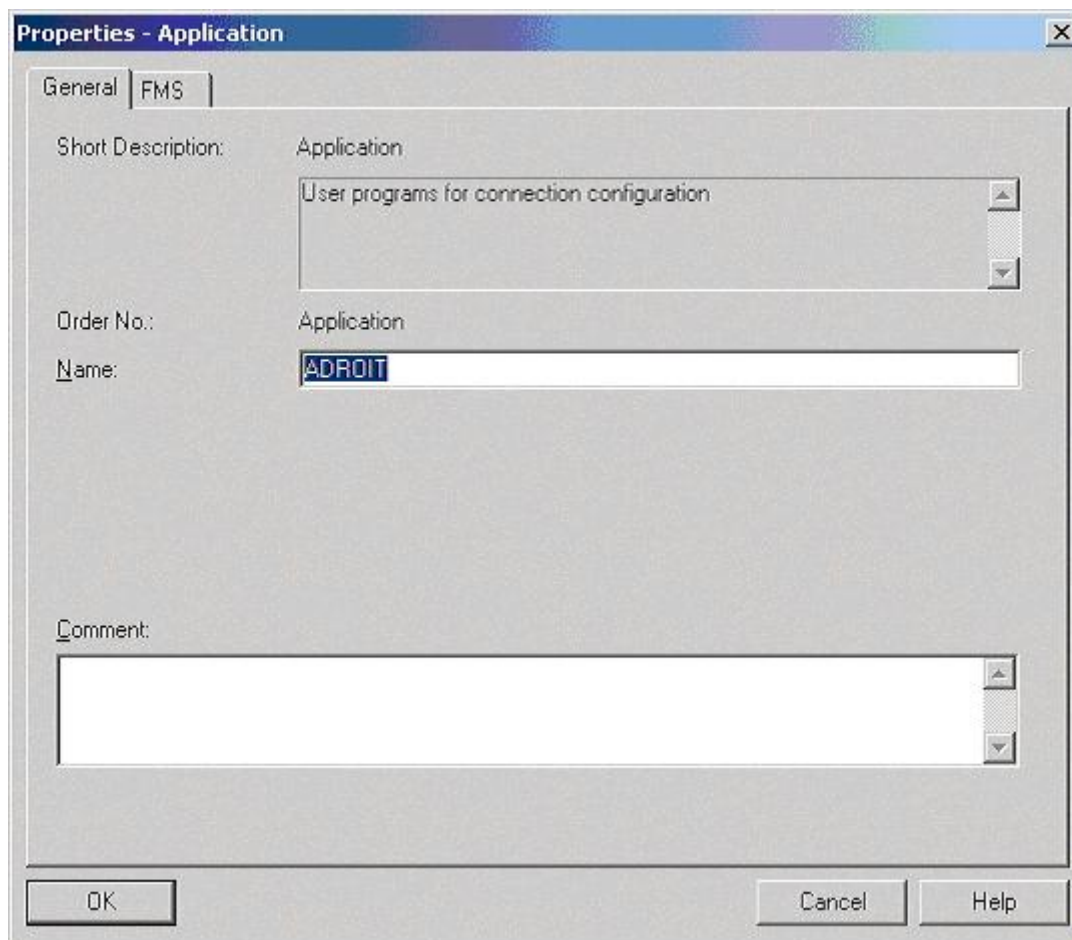


Select “Change hardware configuration”, and click “Finish”.
You will be taken into the station configurator

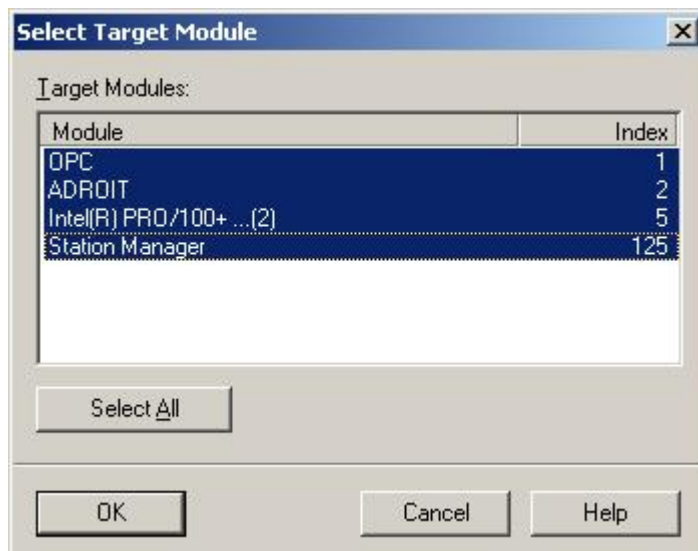


The software will have detected any CP's on your computer during the installation. Profi-bus and other PC cards will be shown here, in this case, our Ethernet adapter appears. We have also added an OPC and Application to our PC configuration. Use your mouse to drag a new application into your PC from **here** to here.

Select the Application, and right-click to get the Object Properties, change the Application name to "ADROIT" all in uppercase.



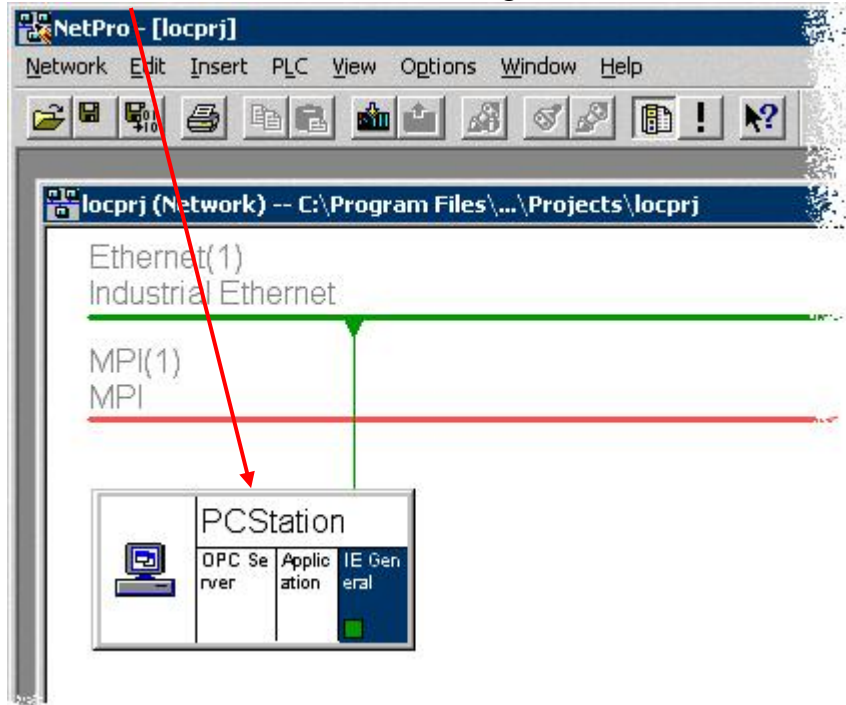
From the Menu select Station/Save and Compile. Compile the station, and then download it as well.



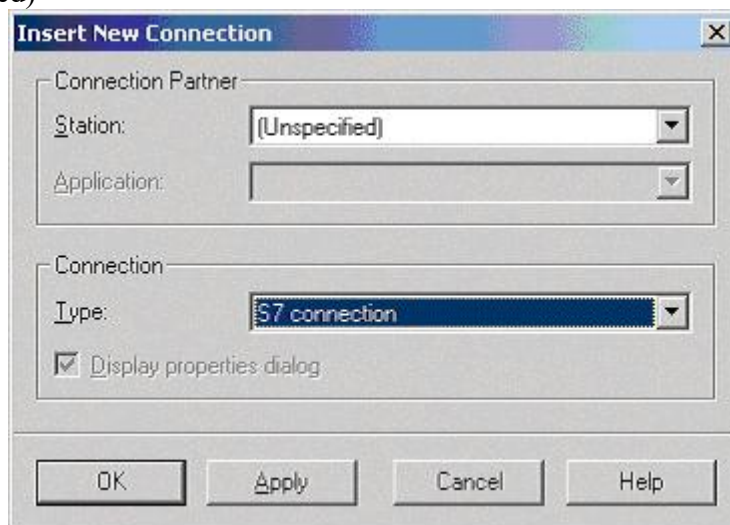
(The dialog shown may differ if using certain Simatic Net versions)
Close the NCM PC Config, the next step is to configure connections.

To set up connections

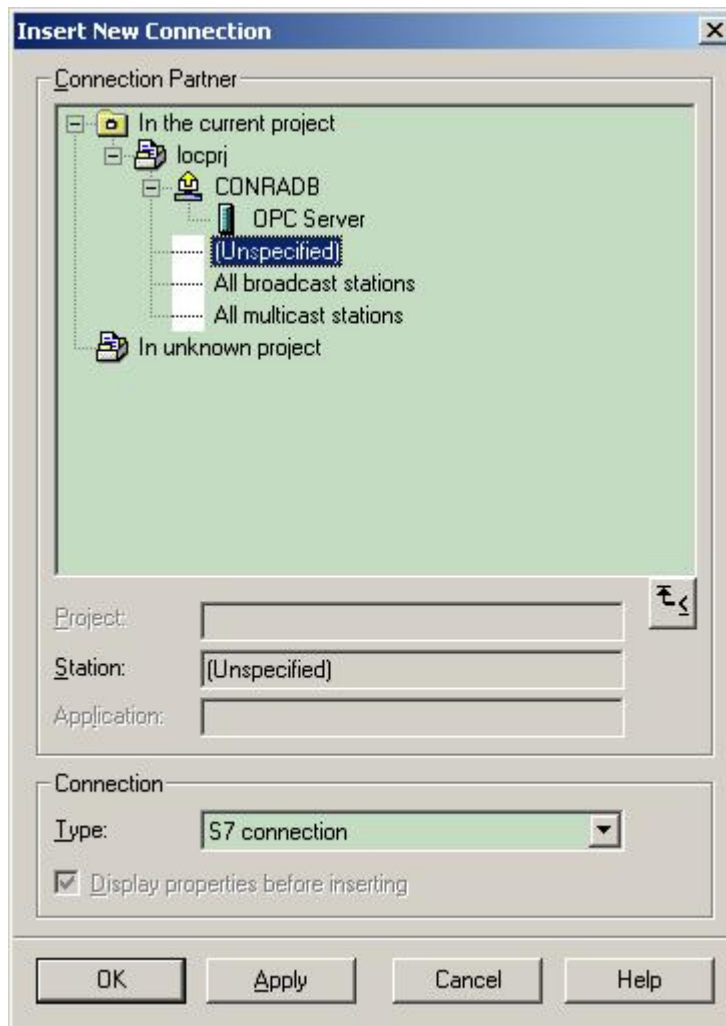
1. Open Setting the PC-PG interface from Control Panel. Ensure that the active interface is "S7-online".
2. Open "SIMATIC NCM PC Wizard"
3. Select "Change local settings" click on Next.
4. Select "Edit network and connection..." Click on Finish, Netpro will launch.
5. Find the PC station box in the diagram.



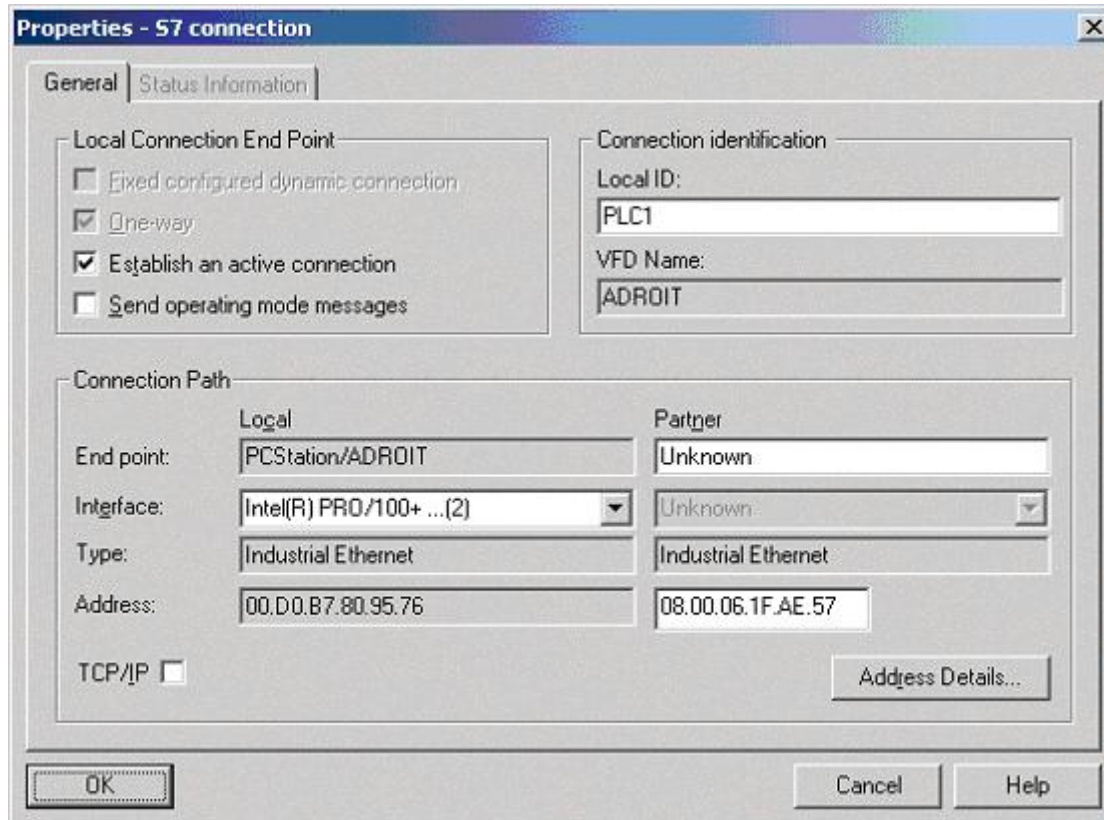
6. Click on Application to select it.
7. Right-click and "Insert new connection"
8. Select "unspecified" for the station.
9. Select "S7 Connection" (Redundant connections probably require S7-RedConnect to be loaded)



For Simatic Net 6.x, the dialog should look like.

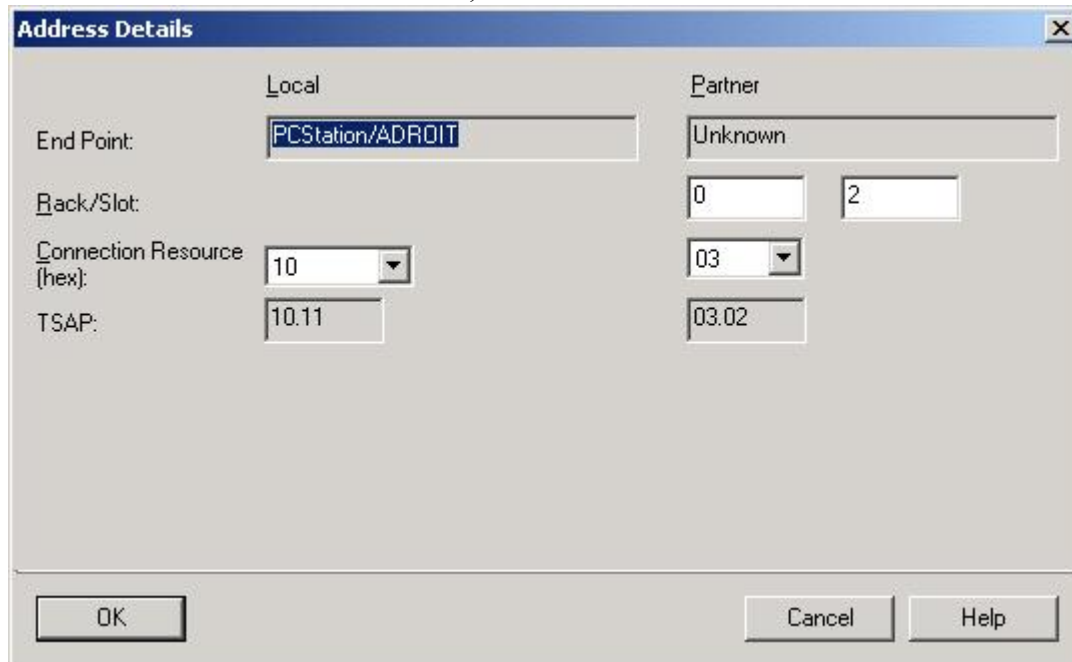


10. Click OK.



11. Type in the PLC address: Depending on the type of network, this dialog and the next will differ, please read the help at this time.

12. Then click on "Address Details", and fill in the details of this connection.



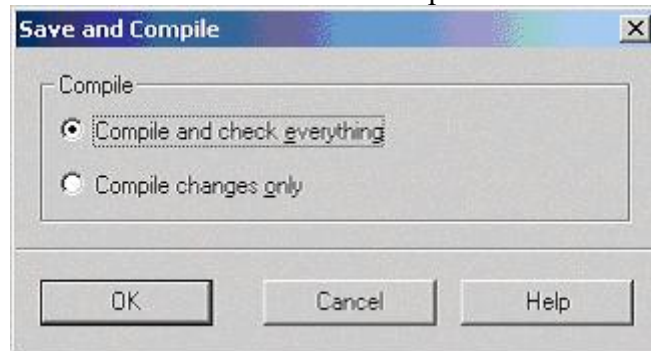
Address details for an MPI connection on CP-5611

13. The connection resource on partner must be 03 as shown in the screenshot.

14. Example if Ethernet or TCP, you need to specify the Rack and position # of the CPU

15. Click OK, Click OK

16. Optionally rename the connection to be the same as the name device agent you want.
17. Perform steps 6..16 for each PLC.
18. Select from the Menu "Network/Save and compile"



19. Select "compile and check everything" click OK.
20. The window that appears should indicate no errors.



21. Click on the Station itself now, (the station, and not the application will be highlighted).
22. Click on the download tool button or the menu "PLC/Download Selected stations.
23. Click OK on any warnings/messages you may get.
24. To download all modules, just click on OK.
25. Exit Netpro.
26. PC and connection configuration are now complete. Open Setting the PG/PC Interface (from control panel), and select the relevant CP from the "Access point of Application" at the top of the dialog. E.g. CP_L2_1:
27. **Reboot the PC** (required after first time, recommended if having problems).

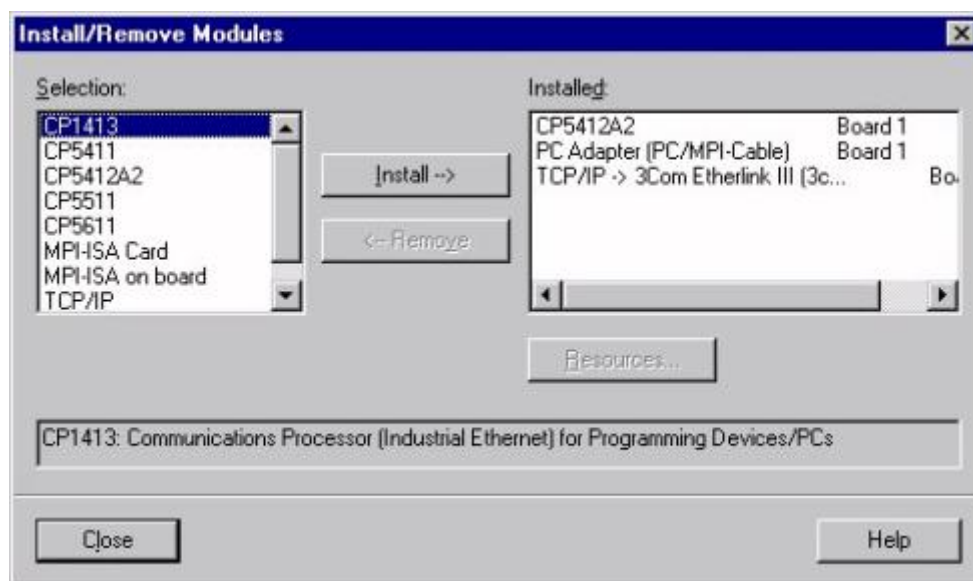
If you want to configure connections again, change the "Access point" back to S7_ONLINE, and change/add connections as above using the Simatic Wizard. Remember to change it back to the desired CP afterwards for the Adroit driver to work.

Continue to the **Device Configuration** section.

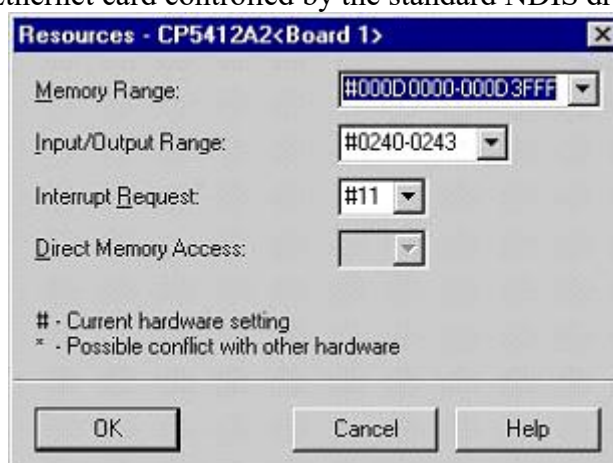
Changing hardware settings (L2)

When using version 4 or later, the physical I/O settings are configured on a separate dialog. To change hardware settings for the local CP with previous versions, follow the instructions at the beginning of this document. If changing the IRQ is required follow the procedure below.

1. Open the "Setting the PC/PG interface" applet in the Control Panel.
2. Select install.

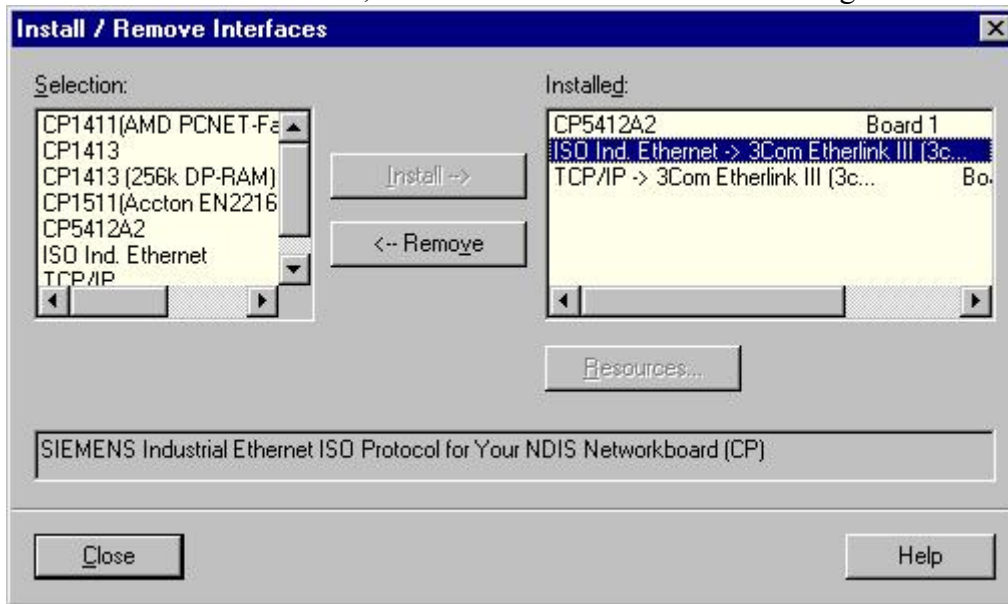


3. From the list on the right, select the relevant card or interface.
4. The “Resources” button will become available.
5. Click on it.
6. Change any configuration settings as required. (This option is not available on H1 because your Ethernet card controlled by the standard NDIS driver.)



H1 Hardware Settings

The H1 resource settings cannot be changed since these are controlled by your card's own NDIS driver, but the screen should look something like this.



Bus Parameters (L2)

The bus parameters used on all devices connected to the L2 Network should be the same. The Sinec reconfiguration program allows 2 automatic and one manual setting for bus parameter calculation. The Standard mode and Universal mode settings automatically calculate bus parameters for you by the program while the user mode allows for the manual setting of bus parameters.

Universal mode is recommended, and will ensure that the parameters closely match those visible in Step 7. Step 7 names for the bus parameters are shown in brackets in the sample dialog below.

SINEC Reconfiguration CP 5412 (A2)

PC hardware parameters

Device name: Interrupt:

I/O port: Dual-port RAM:

Other L2 bus parameters

Call retry counter: X

Slot time (TSL): Bit times

Quiet time (TQUI): Bit times

Setup time (TSET): Bit times

Min. station delay rem.(min.TSDR): Bit times

Max. station delay rem.(max.TSDR): Bit times

Target rotation time (TTR): Bit times

Gap update factor (G): X

L2 bus parameters

L2 address: Mode for L2 bus parameter calculation:

Highest station address:

Number of masters:

Number of slaves:

Transmission rate:

Other Parameters Calculate New OK

CP5412-A2 Bus parameter configuration using Simatic net version 4.

Properties - CP_L2_1

PROFIBUS Network Mode FMS/DP-Protocol S7-Protocol

Station Parameters

Local Station Address:

☐ Not the only master active

☐ PCS7 time synchronization

Network Parameters

Transmission Rate:

Highest Station Address:

Profile:

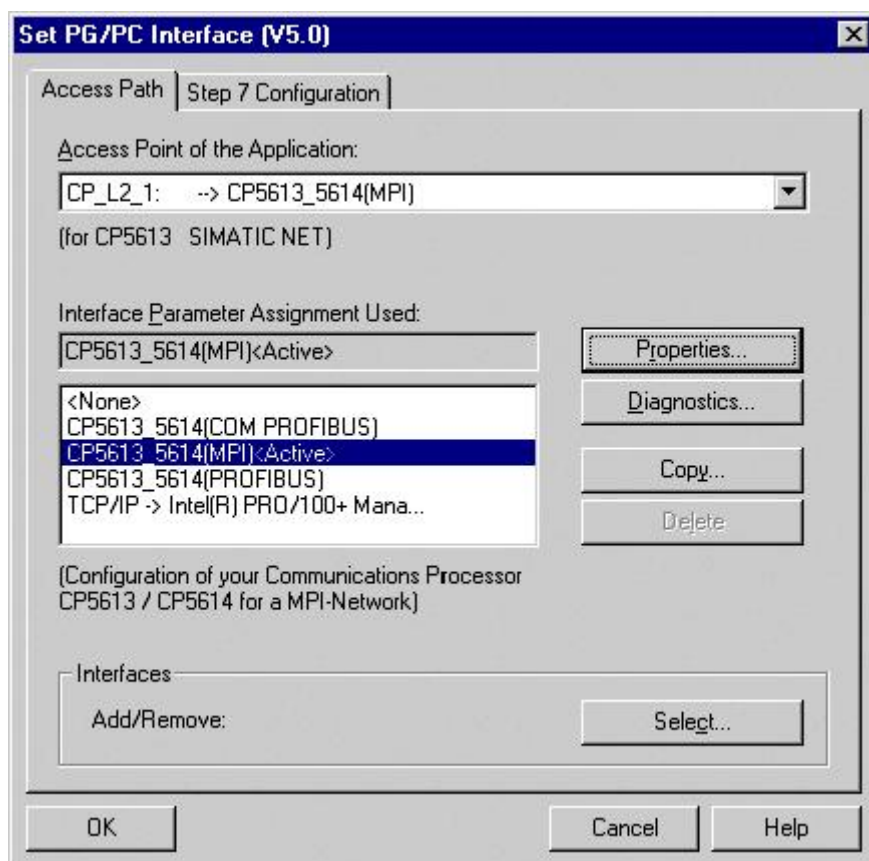
Number of Masters:

Number of Slaves:

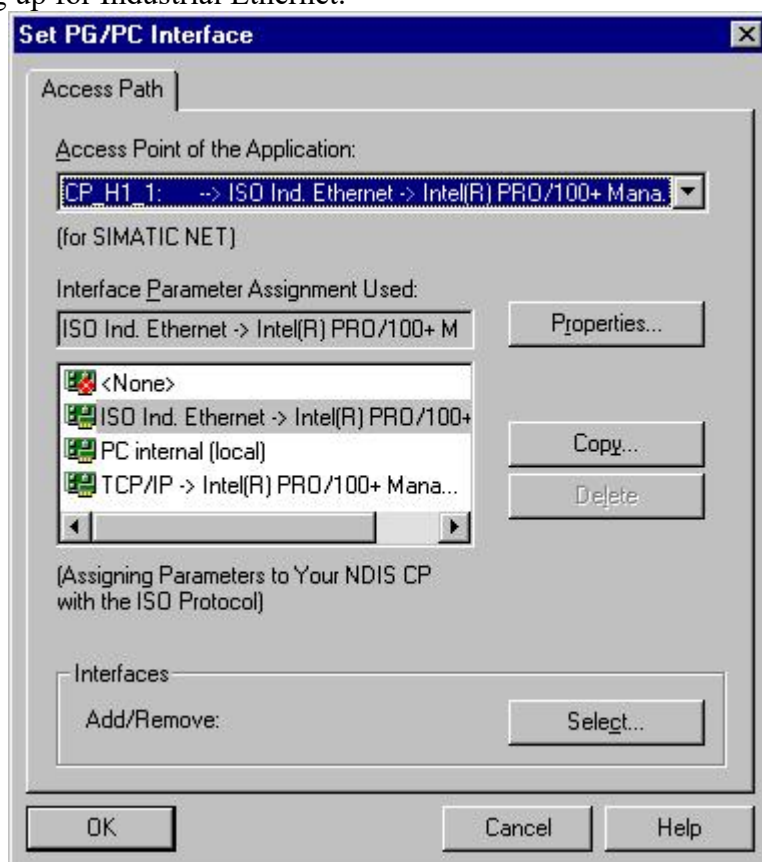
Bus Parameters...

OK Cancel Standard Help

When installing a CP-5613 we set up as:



When setting up for Industrial Ethernet.



Using S7-RedConnect

S7-RedConnect is a software package, which maintains redundant connections to Step-7 PLCs, which have been configured with 'H' system software. The redundant connection is transparent to the PC user, and occurs in conjunction with a CP-1613 interface card connected to the Industrial Ethernet or TCP/IP network.

If using S7-RedConnect please follow the instructions in the SIMATIC readme, you will have to install and use Step7 to configure the redundant connections. This cannot be done in Netpro. The PLCs also need to be correctly configured for redundant connections.

Note: Using Step-7 to configure connections on a machine.

If using Step-7 to configure connections (As we do with S7-RedConnect), do not download the Station as with the PC Configuration wizard, instead compile and save the project. In "Setting the PC-PG interface", Click on the "Step-7" tab and browse your projects' XDB file into the dialog where is requires an XDB. The tab will then show the CP to be used and it's properties.

S7-RedConnect install procedure:

1. Disable Plug-Play in the BIOS.
2. Apply Service pack 2 for Win2K Professional.
3. Turn off the computer, and insert the CP1613.
4. Windows will detect the new card, Install the drivers from the S7-RedConnect CD, do not use the floppy disk.
5. Install S7-RedConnect
6. Install Step-7 Basis; include NCM Ethernet, Authors and Acrobat reader.
7. The PC will re-boot, and continue automatically with the next 2 steps.
8. Install Step-7 Service Pack 2
9. Install Step-7 Hotfix
10. Install H System software.
11. Use Authors to transfer any licenses that were not transferred to the computer during the installation.
12. Copy the "Setting the PC-PG interface" icon onto your desktop. Edit it and remove all of the command line parameters.
13. Copy the "Configuring networks" icon onto your desktop.

Use the Step-7 project as set up on the PG unit as a basis to configure the connections for the CP1613.

1. Copy the S7 project from the PG unit to this station.
2. Run "Setting the PG-PC interface " shortcut, and click on the S7 tab, and browse in the XDB file, which was generated by Step-7 under your project.
3. Add the CP1613 to the workstation in Netpro using Hardware config for the Station.
4. Add an Application, call it "ADROIT " (all uppercase, no quotes). Save and exit the H/Ware Config.
5. Now edit the connections.

6. Create a new connection; the partner must be the 'H' system PLC pair you wish to communicate to that has been set up already. The connection type must be S7 redundant. Change any properties that you want at this time.
7. Create new connections for all other PLCs. (Connection names can be anything.)
8. Download the modified PLCs, as their connections will have changed to accommodate the CP-1613 added to the network. To do this in Netpro, click on each PLC, and then select "PLC/Download/Connections and gateways" in order to re-configure the connections from the CPUs.

Open the Adroit setup.

1. If not already done; install the Adroit S7 Functions driver.
2. Add and configure each device, using the connection names as specified in step 4-5 above.
3. The H system (RedConnect) connection's status can be determined using the Diagnostics program found under the "Simatic/Simatic net/NCM Ethernet" start menu.

Note: This driver does not support a mechanism for determining which 'H'-system path is currently active/productive.

Adroit driver installation

The Adroit driver is installed in the normal fashion using the “Add Driver” button on the Adroit set-up dialog “Drivers” tab.

Device Configuration

The Adroit driver setup will allow you to configure your driver devices. Configure one device per PLC (i.e. no duplicate L2 addresses). More than one device to an address will cause undesirable operation. The set-up dialogue will not check for duplicate addresses, since duplicates are valid, so long as they are not used together in the same agent server. A maximum of 32 devices may be used at one time; this is a limitation of the CP 5412-A2.

Device Configuration Dialog

SIEMENS S7 Functions Protocol Driver : S7_H1

Port Selection

☐ Enable Secondary

☒ Primary

☐ Secondary

Network

Network type: ETHERNET

CP card: CP_H1_1

Remote Details

Address: 08.00.06.01.00.28

Remote TSAP: 03.07

☐ Monitor PLC watchdog

Watchdog address: DB41:DW2

Watchdog period (sec): 10

Only one network type is supported at a time. The Layer 2 address format will depend upon the network type:
 PROFIBUS - station number
 ETHERNET - mac address
 TCP/IP - i.p. address.

For more help read SIES7FN.DOC.

Buttons: OK, Cancel, <<, >>, CP Setup

If an “Advanced PC Configuration detected” message appears, the dialog will be different, some settings will not be required.

Port Selection	Disabled
Network Type	Select PROFIBUS/ETHERNET/TCP-IP
CP 5412	Select the local CP card. Select the device name for the Access point.
Address	Address of the partner (station): PROFIBUS: this will be a PLC's L2 station address 1..127 ETHERNET: this will be the MAC address of the remote card e.g. 01.00.06.00.07.00.20 TCP/IP: This will be the IP address of the remote card e.g.

	40.10.10.10
Remote TSAP	This is divided into 2 hexadecimal numbers. First is the device number 01=PG, 02=OS, 03=Other (Adroit) then after the period comes the rack and slot. v bits 7..5 = rack (subsystem) v bits 4..0 = slot e.g. 03.02 = Non S7 or other, with rack zero slot 2. The slot number is the starting slot number in which the CPU resides . Slots are numbered from the left starting at one. Depending on the model of CPU, different slot rules may apply, see relevant Simatic documentation for particulars.
Monitor PLC watchdog	Enable the watchdog facility (default = off)
Watchdog address	A register or word address where the watchdog variable is found.
Watchdog Period	The duration in seconds between each watchdog test done by the driver.
<< >>	Browse previous / next device
CP Setup	The Sinec setup program can be run directly from this button.

A RSAP is not required, since 00.00 is assumed. When assigning L2 BUS addresses; remember that an unused address must always be reserved between stations on the BUS. Only use either odd or even addresses. Address zero should also not be used, since it is reserved for a PG or diagnostic node. If using MPI, the remote tsap to use may be 01.00.

Watchdog configuration

Watchdog configuration is optional, and is intended to allow detection of the started /stopped state of the PLC. Without the watchdog a user has no way of knowing that the remote CPU has gone into stop mode, since the communications processor, being separate, continues to function as normal. If not required the watchdog checking should be disabled.

Watchdog operation

If enabled the watchdog operates by flagging the device as off-line, almost as if there was a cable fault. The driver then polls the configured 16-bit word at 10-second intervals. If the PLC variable's value is zero, the driver flags the device as on-line, and overwrites the zero with a one. If the PLC variable's value is not zero, the driver will re-test the variable 2 more times, before flagging the device as off-line. The extra 3 tries constitute a grace period, after that all scanning will be bad until the watchdog variable is zeroed. The driver continues to poll during this time.

Watchdog PLC coding

The simplest PLC code should simply zero the watchdog variable on every cycle. The watchdog may also be used to detect the presence of the SCADA, by doing the watchdog in an OB executed every 5 seconds. The OB must test the variable before setting it to zero, if the test fails 4 times in a row, it can be assumed that the Adroit

driver is no longer setting it to one. The PLC may then be programmed to take some action, on the assumption that there is no longer any operator control.

STOP

Note! If not using the Advanced PC Configuration SINEC. Go back to the Sinec setup (Setting the PC-PG interface) program and select the Adroit database file (ADROIT.LDB) in the database field of the dialog, and then re-boot the machine for the last time. You need only ever do this once (step not required in Simatic NET 6.x onwards).

How it works (Simatic.NET versions < 6.0 only).

This paragraph is optional information provided here for extra reading only.

The Adroit driver setup dialogue creates a text database file

C:\SINEC\COMS7.NT\ADROIT.TXT that can be edited using the COML-S7 utility.

Saving over this file using the Sinec program is not recommended because it would then get out of step with the driver's internal copy of this configuration data. The text database can then be converted to a binary LDB file called ADROIT.LDB, by running the database conversion utility RCS7.EXE. All of this is done for you automatically by Adroit driver set-up when clicking on the OK button in the SIEMENS S7

Functions Protocol Driver configuration dialog. If the Advanced PC configuration or Step-7 is loaded on the computer, the COML-S7 utility is not present, and the driver connections must be configured and downloaded, the database then lives in an XDB file. This is detected by the presence of a registry key

"LOCAL_MACHINE\SOFTWARE\SIEMENS\NCMS7\KvDI1".

Advanced PC Configuration dialog

SIEMENS S7 Functions Protocol Driver : PLC101

Port Selection

☐ Enable Secondary

☒ Primary

☐ Secondary

Network

☒ Using Simatic v6.3 and Higher

☒ Allow the use of multiple CP cards

CP card: CP1613(RFC1006)

Connection: PLC101

Remote Details

Open NCM/S7 Project...

PDU resources: 114 words

☐ Monitor PLC watchdog

Watchdog address: DB41:DW2

Watchdog period (sec): 10

Advanced PC configuration detected.

PLC connections must be configured in the PC Configuration Wizard.

If you are using S7 RedConnect, please configure connections via Step-7.

OK, Cancel, <<, >>, PC/PG Interface..., Help

Allow the use of multiple CP cards

Option is only available if using Advanced PC configuration Simatic Net versions. Enable to allow different devices to select other CP cards.

CP card

Using Simatic v6.3 and Higher

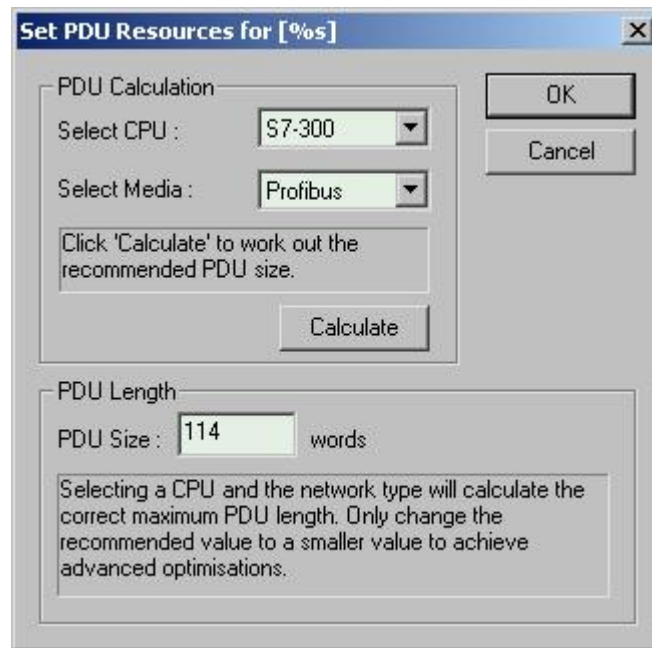
Connection

Select your CP card from this list.

Option is for when using Simatic Net versions 6.3 and higher.

Select your connection name from this list. The connection name may be the same as the device name.

PDU size dialog



It is possible to either have the driver calculate the correct PDU length, or type in your own value. The PDU length represents the maximum number of words that the driver should request from the remote CP. If you receive a event-message that indicates a PDU is too large, this value should be reduced, and the Server re-started for the change to take effect.

If the driver attempts to scan a PDU that is too long to handle in the remote CP, the driver actually takes the entire scan-task offline. This is so that we does not keep on reporting the same error each time in successive attempts.

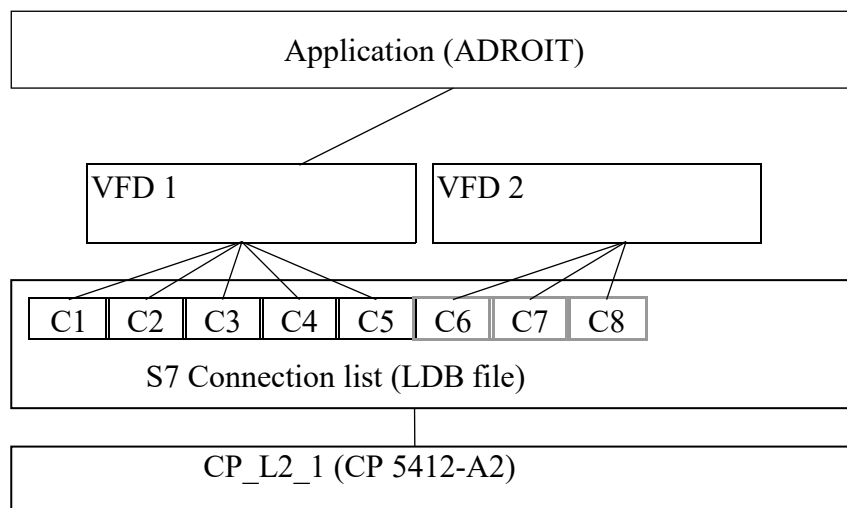
This option-dialog replaces the registry setting in driver versions prior to driverRevision 30.

Scanning configuration

Scanning configured points should be optimized to fit into blocks of up to or less than 114 words depending on media type (see: registry section and PDU dialog section for more). Scan points should be grouped according to desired scanning speed and also by data type.

VFDs in Simatic Net versions prior to 4.0

Before the advent of the “Advanced PC Configuration” Simatic net V4.0, which uses Net-pro to set up networks, each device configured gets assigned as a connection on a VFD in the SIEMENS connection database .LDB file. One VFD is used by the driver for all devices.



Adroit set-up configures each device as a connection on a single VFD.

Supported addresses

The format of the address string is as follows:

DB ww :DW xx[.yy] [z]

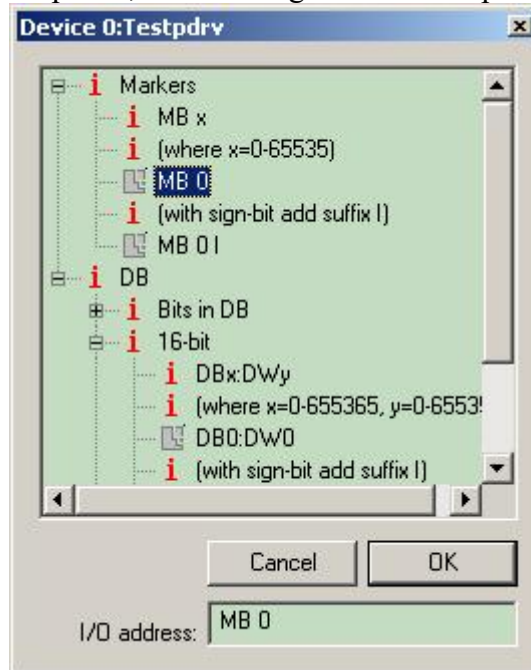
where

DB is a prefix or mnemonic for data block, ‘ww’ is the S7 data block number. DW or DD refer to single or double data items respectively, and ‘xx’ is the offset. Optionally a bit offset ‘[.yy]’ may appear followed by an optional data type suffix ‘[z]’ as below.

Device	Max Range	BOOLEAN	INTEGER	REAL
Digitals	DB 65535:DBX65535.7	DB 255:DBX255.7		
BYTE (marker)	MB 65535	MB0.0	MB 0	MB 0
CHAR (marker)	MB 65535 I		MB 0	MB 0
BYTE	DB 65535:DB65535		DB 0:DB0	DB 0:DB0
CHAR	DB 65535:DB65535 I		DB 0:DB0	DB 0:DB0
WORD	DB 65535:DW65534		DB 0:DW0	DB 0:DW0
INTEGER	DB 65535:DW65534 I		DB 0:DW0 I	DB 0:DW0 I
DWORD	DB 65535:DD65532		DB 0:DD0	DB 0:DD0
LONG	DB 65535:DD65532 L		DB 0:DD0 L	DB 0:DD0 L
REAL	DB 65535:DD R			DB 0:DD0 R
STRING	DB 65535:DB6540 S80	DB 100:DB50 S20		

Strings: The string format fills the remainder of the string with spaces (character 32) up to the length specified. E.g. the address “DB 100:DB50 S20” allocates 20 bytes starting at byte 50 to the string. If the String presented in the SCADA is longer than 80 bytes or the length (e.g. 20), then it is truncated before being written to the PLC. The spaces are not trimmed when presenting the PLC value to the SCADA.

When using the Agent Explorer, the scanning browser will provide hints:



Note: If after selecting a device, if you do not see a default address appear in the “Source Address” field of the scanning dialog, the driver has not loaded. Look at the event viewer to establish the reason.

Message protocol

The SIEMENS message protocols are not shown in Datascope diagnostic displays. The driver data-scope screens display a running commentary on all operations performed instead, and display the data sent and retrieved.

PLC Software

No additional software is required, except for the correct configuration of the L2 card. Each PLC must be set up to have a non-S7 partner (i.e. Adroit) in its project. No connections need to be configured. All other default settings can be used regarding bus parameters. Using default parameter settings in Sinec set-up will match those in Step 7 for your PLC.

Using Step-7

When using Step-7 to project PLC networking, (I.e. Step-7 is on the Same PC as Adroit) the driver will use [Advanced PC configuration](#) mode. Herewith follows the quick guide to Step-7 connection projecting:

1. Open your project
2. Open the network connections, and add a Simatic workstation to your network.
3. Add in the same adapter you are using to the workstation. (If using a standard Ethernet card, this will be “IE General”).
4. Add an application object to the station, and call it ADROIT, all in uppercase.

5. Double-click to add and configure the PLC connections, to see how, go to “[To setup connections](#)”
6. Save and compile connections.
7. Click on the PC Station, and download, this will save the PC connection settings into an XDB file, which is used by the driver.
8. Open Setting the PC-PG interface, click on the “Step 7 Configuration” tab, browse in the folder where your project is stored, in there you will find an XDB file.
9. Continue driver configuration as in [Advanced PC dialog](#).

Performance

Performance Counters

The S7 Functions has 2 performance counters in the performance registry that differs from the standard counters provided by adroit drivers. The counter and per second counter described are simple activity indicators.

S7 poll count	Total number of s7 polls done.
S7 SAPI polls per second	Polls per second, this drops as more data is transferred.

Throughput

L2: The throughput of the driver is documented only as an approximate guideline to determine whether the system is suitable for the intended application, and may change without notice.

Tested at 1.5MBaud using long and short telegrams

PLCS	Transaction/sec
1	10 to 11
2	20
3	25
12	65

Tested at 187.5KBaud using long and short telegrams

PLCS	Transaction/sec
1	4
4	26

Maximum transaction length is 114 words. The above figures were derived using a typical mix of 5 telegrams average 80 words. Telegram lengths below 10 words were found to give higher transaction speeds, but less total throughput.

H1: Achieved 24 Transactions/sec and 1300 registers/sec to 1 PLC, not tested on multiple PLCs.

Simatic.Net Compatibility

Any driver revision is backward compatible with Simatic.Net but not forward compatible. Here is the Driver revisions that is compatible with the various Simatic.Net versions:

Driver revisions 1 – 30 supports any Simatic.Net versions up to and including Simatic.Net v6.2!

Driver revisions 31 – 33 supports any Simatic.Net versions up to and including Simatic.Net v6.3!

Driver Revision 34 supports any Simatic.Net versions up to and including Simatic.Net Software Edition 2008!

Troubleshooting

While every attempt has been made to ensure technical accuracy, different software and hardware versions may operate in an undesired/unexpected fashion. We thus welcome any comments and suggestions, and will include any pertinent technical data submitted.

I cannot see the driver in the available list in Setup

The driver will only appear in the list, once the simatic networking API files (SPIAPI.DLL etc) from Siemens have been installed. Follow the Simatic installation procedure.

License required message appearing

You will need to acquire one of the licenses described in this message to operate this package. Please contact your siemens representative and quote the GK number and package description. The message will appear on loading and at other times while using the package. Our driver will still function correctly, if you acknowledge this message each time it appears.

If Using “Advanced PC configuration”

Ensure that your Application name is “ADROIT”.

If you download a new configuration to this PC, return to the driver setup, and verify your settings by clicking OK in the driver setup. Re-start the agent server to ensure proper operation.

Restarting

If Adroit driver configuration is changed, the Agent server must be re-started to accept the new changes. Only re-boot if you change parameters within the SINEC set-up and the save dialogue tells you that a reboot is required to put the changes into effect. I.e. bus parameters changed. If using the version 4 of SIMATIC NET software, re-booting the PC may always not be required when changing settings, use the reset and restart feature in the PG-CP setup program. If bus address conflicts are suspected shut down

the computer and then leave the machine switched off for about 10 seconds before re-starting.

Another application is using the CP

It is not possible to run any of the CP diagnostic features or configure the CP settings and connections while either Step-7 or Adroit is running. You will have to close these applications first. In many cases (depending on S7 network configuration) it is not possible to run Step-7 and Adroit at the same time.

Diagnostic file

The driver creates a diagnostic file extension .DIA with the same name as the device agent, containing all major communications and other events. Delete this file if it becomes too large, or disable it by setting the registry value "UseLog" in the device configuration (accessible using regedit) to zero. A value of '1' for this value will re-enable it. The default setting is disabled, since this information also appears in the Error Log screen of the protocol driver monitor. For more Advanced registry settings see "[Registry Keys](#)"

Diagnostic Tools (L2)

If using Advanced PC configuration, select Setting the PG/PC Interface and then select your CP and click on Diagnostics, skip to step 6.

A diagnostic tool called CPMONNT.EXE is provided as part of the SIEMENS SINEC installation. It can be used to determine which nodes are participating on the bus and also to display their L2 addresses. After connecting to a CP this program has a life-list feature to see which stations are receiving the token.

To create a shortcut to the program:

1. Right click on the Start button.
 1. Select Open All Users
 1. Double click on Simatic Folder
 1. Double click on SIMATIC NET folder
 1. Select File\New\Shortcut
 1. Click on Browse
1. Find the file "C:\Sinec\tools.nt\cpmonnt.exe"
8. Click on Next
9. Click on Finish.

To find what stations are active on the network.

10. Run the program.
11. From Option select "Select CP"
12. Click on "CP_L2_1:"
13. Click on OK
14. From Command select "Update Lifelist Remote"
15. A dialogue with 126 tick-boxes will appear. If the cable is terminated correctly and the card is working, you will see at least one station is active, the PC itself.
16. The stations shown can be used as station numbers in the Adroit driver set-up dialogues or connection configurations.

Diagnostic tools (L2) Advanced PC

Run the “Configuration Console” shortcut, select your profibus module, select the option “Bus nodes”, this option is not present for H1/Ethernet CPs.

Other errors

Invalid DB address errors occur if an attempt is made to scan a non-existent offset within a DB. To fix this, un-scan the offending point, save your database (WGP file) and re-start the AS to get rid of the error.

You may continue to operate the system without re-starting the AS, however all tags in the affected scan task will remain bad (not scanned) until you re-start.

Not connecting

If the driver fails to connect and produces an error message containing: “Invalid CREF or connection” then you need to run the Set-the-PC-PG program, and set the application access point to be the same as the CP being used, E.g. CP_L2_1:

After upgrading the driver stopped working

Open Adroit Setup, click on drivers, and configure any one of the devices. Check that your settings are unchanged and still correct. Click on OK to allow the driver to save any new settings the upgrade may have introduced.

Advanced PC detected incorrectly

If the driver starts using advanced PC config, because it has detected a Step-7 installation, please see the registry key **COMLS7** in the Appendix [Registry Keys](#), this value must be added and set to 1.

Removing a step-7 installation from your computer completely

From a technical note :” **Complete deinstallation of STEP 7 including all software packages**”

QUESTION:

How can I completely deinstall STEP 7?

ANSWER:

We recommend that you always deinstall **all** Siemens software and reinstall it after deinstalling STEP 7. Otherwise there might be problems with the Siemens products still available.

The procedure for a complete deinstallation is described in the following:

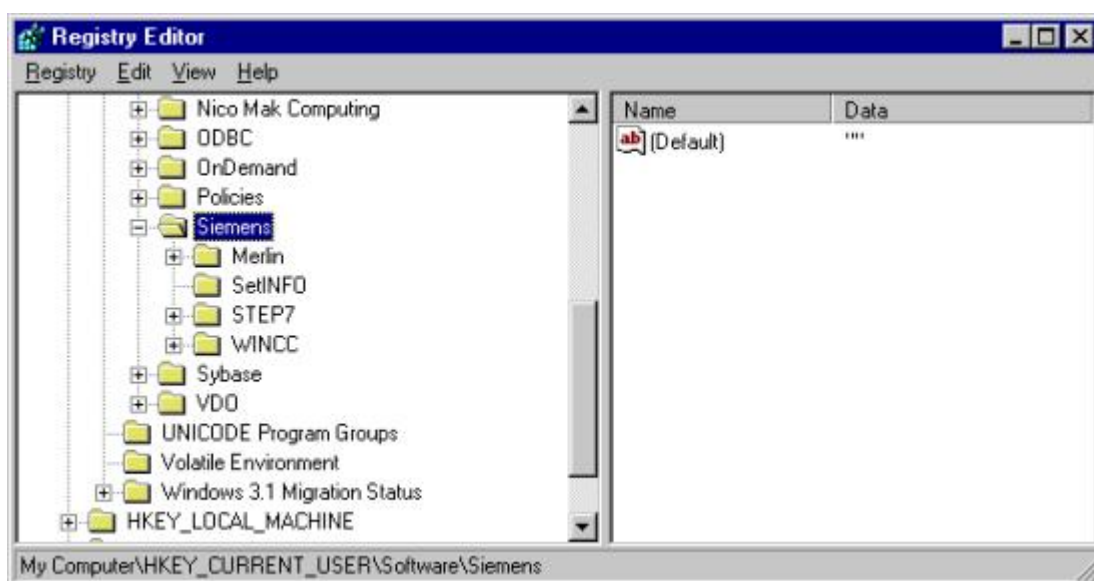
Warning:

In the deinstallation described the complete SIEMENS folder and all Registry entries of the Siemens software installed on your computer are deleted. If this folder still contains projects, then you must copy them to another directory using the SIMATIC Manager function "**File > Save As&**". You can also save the data in another directory or on a different drive using the function "**File > Archive**". Do **not** copy or move the projects using the Windows Explorer. Still save your GSD files from the directory "<<DRIVE:>>\Siemens\S7data\gsd\".

1. Start the "**Software**" via "**Start > Settings > Control Panel**" and deinstall all the SIMATIC applications.
2. Then restart the PG / PC.
3. Start the Registry editor via the menu command "**Start > Run**".

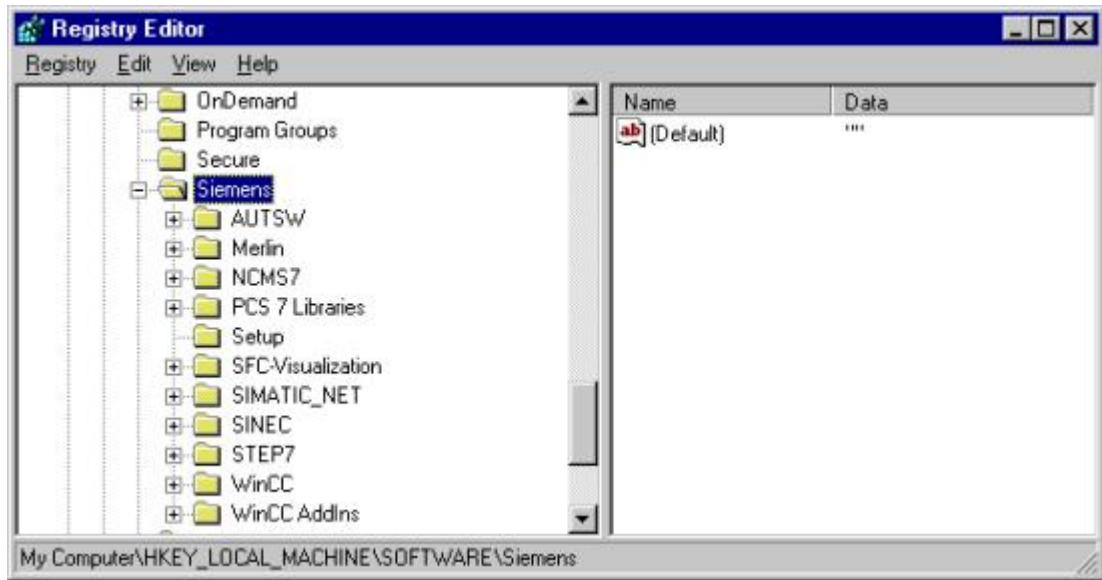


4. Enter **"regedit"** and acknowledge with **"OK"**. The Registry editor now starts.
5. Open the directory **"HKEY_CURRENT_USER > Software"**. There you will find the **"SIEMENS"** subdirectory.



6. Select this directory and delete it.

7. Open the directory "HKEY_LOCAL_MACHINE > SOFTWARE". In it you will find another "SIEMENS" directory.



8. Select this directory too and delete it. Then the Registry is free of all SIEMENS entries.
9. Close the Registry editor and then delete the complete Siemens directory (<<DRIVE:>>\Siemens) as well as all the S7 files in the Windows system directory. For this you open the Search mask via "Start > Search > Files/Folders..." and search for all the "S7*.*)" files.
10. Via "Start > Search > Files/Folders..." search for the "citamis.str" file in the Windows directory (e.g. C:\WinNT or C:\Win98) and delete this file.
11. Then delete the Recycle Bin.
12. This then completes deinstallation.

Using a CP5611

Remove all Siemens components, using above procedure (not needed if all you have is Simatic-Net prior to 07/2001). Install from the "Simatic-Net 07/2001" CD, select all of the components. A re-boot may be required ½ way through the install. Continue as normal. (Amongst, but not limited to, the symptoms encountered prior to doing this are the Adroit setup program crashing when you try to access the driver configuration dialogs.)

The CP 5611 drivers may halt the system if used on a dual-pentium system, please consult the simatic-net readme for updates.

What is a connection, why do I have to save and download?

Adroit driver



Simatic API
(spiapi.dll)



Simatic Windows

Drivers



CP card

At this point, the organization of the software needs to be explained so we can better understand how to troubleshoot any problems. At some points there are mechanisms that allow us to test assumptions and get connections debugged and working.

The CP card may be a CP5613, or your existing Ethernet adapter, a Simatic net driver is installed in the case of Ethernet that uses your existing windows Ethernet driver. The Simatic drivers and the API together manage connections configured in a database, the database is roughly organized to allow many different applications (programs) to communicate to PLCs without interfering with each other. At the lowest level, the database holds a CP card.

#N CP cards



#N Connections



#N Applications

This means that each application will have a number of connections, each connection will be associated with a single CP. In Adroit, we associate a connection with a device, since normally you will sue one connection per PLC

you need to communicate to. Nothing however prevents the configuration of 2 connections whose parameters are for the same PLC. Adroit will still see them as separate devices. Applications are differentiated by their names. “OPC Server” is an example of an application with no connections. An application thus only needs to know the connection name, the connection uses a specific set of parameters, and a CP, all of which are hidden from the application, allowing the user flexibility to use new kinds of CP cards and connection parameters freely.

So all of this information lies in a database, *when you “download” you are actually updating the local connection database*, which gets used by any applications (the Adroit driver).

Connection appears twice?

Connections will sometimes appear under 2 CPs in our driver, that is because a connection on a CP that has multiple names is still the same CP, Example a CP_5613 is equivalent to the same CP_L2_1:, which is the same in some cases to MPI. Selecting either CP will work, since the CP parameters are all actually the same, and so are the connection parameters.

MPI

If using a CP 5613 to connect to the MPI port of a PLC, you must select MPI for the card, you will know you have done this correctly if you cannot specify any profibus parameters for the CP, besides the baud-rate. The default rate is 187.5 Kbs.

Appendices

Abbreviations and Terms

Application	PC station module for PC Applications and drivers.
Connection	A physical connection point with an address on the L2, H1 or other networks.
CP	Communication Partner <i>or</i> Communications card
FDL	Fieldbus Data-link Layer
L2	Layer 2 Protocol

MPI	Multi-point Interface
OPC	OLE for Process control
PLC	Programmable Logic Controller. Used when referring to a S7 Process Control Unit
RedConnect	SIMATIC software to maintain and manage redundant connections
RSAP	Receive Service Access Point
TSAP	Transmit Service Access Point
VFD	Virtual Field Device (a VFD may have multiple connections, not necessarily to the same PLC)

Registry keys

Global keys affecting all devices:

COMLS7:

Under the reg. key

“HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\SIES7FN”, the DWORD value “UseCOMLS7” must be 1 to force the driver to use the COMLS7 database conversion utility. This value does not exist by default.

PDU sizes:

Under the reg. key

“HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\SIES7FN”, the DWORD value “PDULength” may be used to change the PDU size from 114 words, (S7-400 on Ind. Ethernet allows values up to 238). This value does exist by default. (Default = 114) *This is a dialog setting in DriverRevision 30 onwards.*

Scan mappings to CSV:

Under the reg. key

“HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\SIES7FN”, the DWORD value “ExportScans” may be set to non-zero values to enable the export of a devices scan task start-end to CSV, the file can be used to determine the coverage and overlap or fragmentation of scan-tasks. Export occurs when the device is started. This value takes effect immediately. This value does exist by default. (Default value = 0)

Global keys affecting specified devices:

TXT log-file output:

Under the reg key

“HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\SIES7FN\mydevice”, where “mydevice” is a device agent, the value “UseLog” must be 0 (zero) a value of 1 will enable communication logging to text file.

Connection backoff:

Under the reg key

“HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\SIES7FN\mydevice”, where “mydevice” is a device agent, the value “BackoffTime” holds the time (default 7 seconds) that the driver will wait for after a connection breaks, before it tries to restore/make another

connection. This delay allows the PLC to close it's connection, and for any local resources to be released.

Revisions	
Date	Changes
29-Oct-97	First complete revision
17-Nov-97	Bus parameter notes added to "Driver installation and setup"
18-Nov-97	Old driver disabling/upgrading section
28-Nov-97	Watchdog functionality added
14-May-98	New installation procedures for sinec version4
27-May-98	Procedure for changing h/ware config with version 4 software
30-Nov-99	Updated driver setup screen-shot and explained the use of H1 (Ethernet/TCP) feature just added.
10-Mar-00	String support.
02-Nov-01	MPI support added; SIMATIC installation instructions amended.
14-Jan-02	Set the "Application access point". Compatibility notes added.
07-Mar-02	Added MB and DB byte addresses, fixed the address table, up to byte 65535 supported.
05-Apr-02	Advanced PC Configuration setup is covered. Driver revision 16 onwards required.
02-May-02	S7-RedConnect info and steps added.
12-May-02	Driver does not appear until Simatic-net is installed notice now in bold.
22-May-02	Document updated with Red Connect installation guide steps.
05-Jun-02	Troubleshooter (de-install routine added).
14-Aug-02	Added Advanced PC step on Setting-PC/PG to select S7_ONLINE and back again while configuring.
26-Aug-02	How to force use of COMLS7 utility (Driver rev. 30 onwards).
01-Oct-02	Added De-installation note, and CP6511 note. Back-off described.
22-Oct-02	When setting up connections, resource must always be 03
07-Feb-02	New Reg. keys added : 'ExportScans' and 'PDULength'.
04-Apr-03	Step-7 guide added, doc bookmarks used, contents moved up.
10-Jul-03	Simatic 6.x screenshots added, multiple CPs supported on rev 29 onwards + scan-browser added.
22-Jan-04	Architecture better explained, Connections explained.
23-Mar-07	Added Using Simatic Net v6.3 Checkbox as it solves issues regarding using 2 CP's at a time on v6.3 and onwards. Driver revision up to rev 32.
14-Dec-09	Added Simatic.Net Compatibility List